CASE 01 · BUSINESS ANALYST PORTFOLIO PROJECT

The $1.67M Churn Problem

A SQL + Power BI retention case study, built on the real IBM Telco Customer Churn dataset (7,043 customers).

Role simulated: Business Analyst | Tools: SQL (SQLite), Power BI / Tableau, Python | Dataset: IBM Telco Customer Churn (public)

Source: raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d

26.5%	57.5%	$1.67M	2.2x
Overall churn rate (1,869 of 7,043 customers)	Churn rate in the highest-risk segment	Annualized revenue currently at risk	How much more the risk segment churns vs. average

The business problem

Leadership at a telecom-style subscription business can see that revenue is leaking to churn, but not which customers are leaving or why. Before opening a BI tool, this project starts the way a real engagement would: with a one-page BRD scoping the question, so the eventual dashboard answers something leadership actually asked for.

BRD excerpt (simulated stakeholder: VP, Customer Success)

Field	Content
Business objective	Identify the top drivers of customer churn and quantify revenue at risk.
In-scope questions	Which contract types, tenure bands, and service bundles carry the highest churn risk?
Success metric	Identify the top 3 churn drivers and the customer segment they concentrate in.
Out of scope	Predictive modeling / propensity scoring (a follow-on phase, not this analysis).

Method

The dataset was loaded into a real SQL database (SQLite) rather than opened directly in a charting tool — 10 queries drive every figure in this report and ship alongside it as **churn_analysis.sql**. The FRS below was written before any chart existed, so the dashboard build (Power BI or Tableau Public — a 15-minute rebuild from the same query output) answers the BRD's questions rather than whatever the data happened to make easy to plot.

Dashboard requirements (FRS excerpt)

• Executive page: overall churn rate, revenue at risk, and the top-3 driver segments as headline tiles.

• Drill-down page: churn rate by contract type, tenure bucket, payment method, and support-bundle status.

• Filters: contract type, internet service type, tenure band.

• One calculated field: revenue-at-risk = SUM(MonthlyCharges) WHERE Churn = 'Yes', sliceable by segment.

Key findings

Contract type is the single strongest lever: month-to-month customers churn at 42.7%, against 11.3% for one-year and 2.8% for two-year contracts. Tenure compounds it — churn is highest in a customer's first year and falls steadily after.

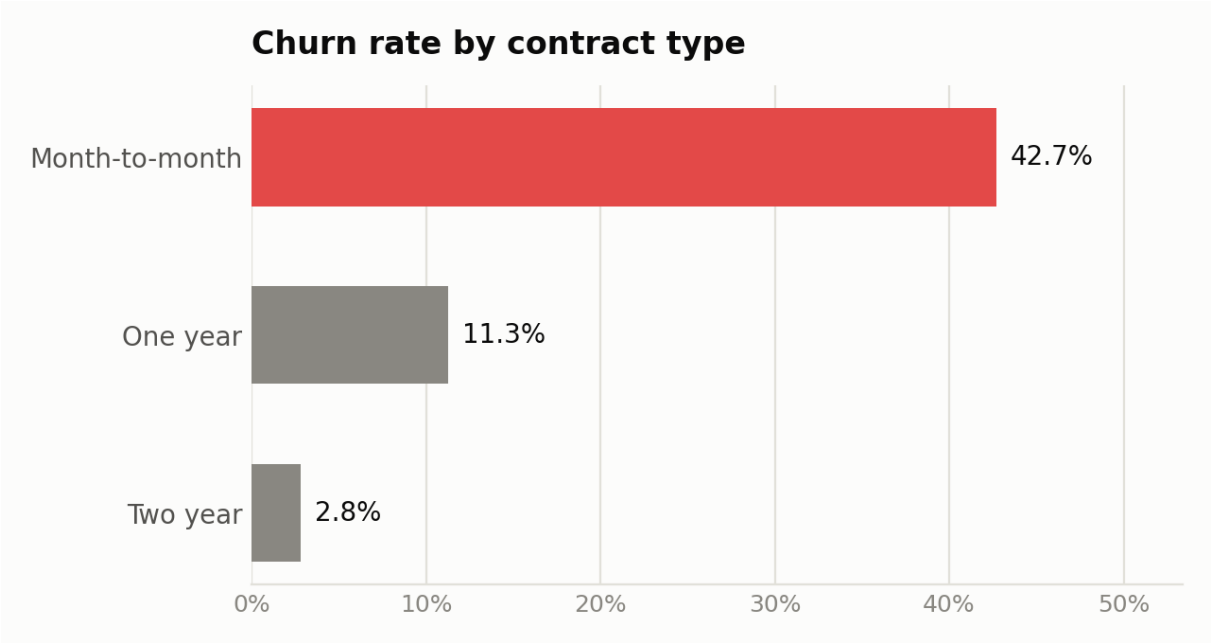


Fig. 1 — Query 2: churn rate by contract type (SQL: churn_analysis.sql, query 2).

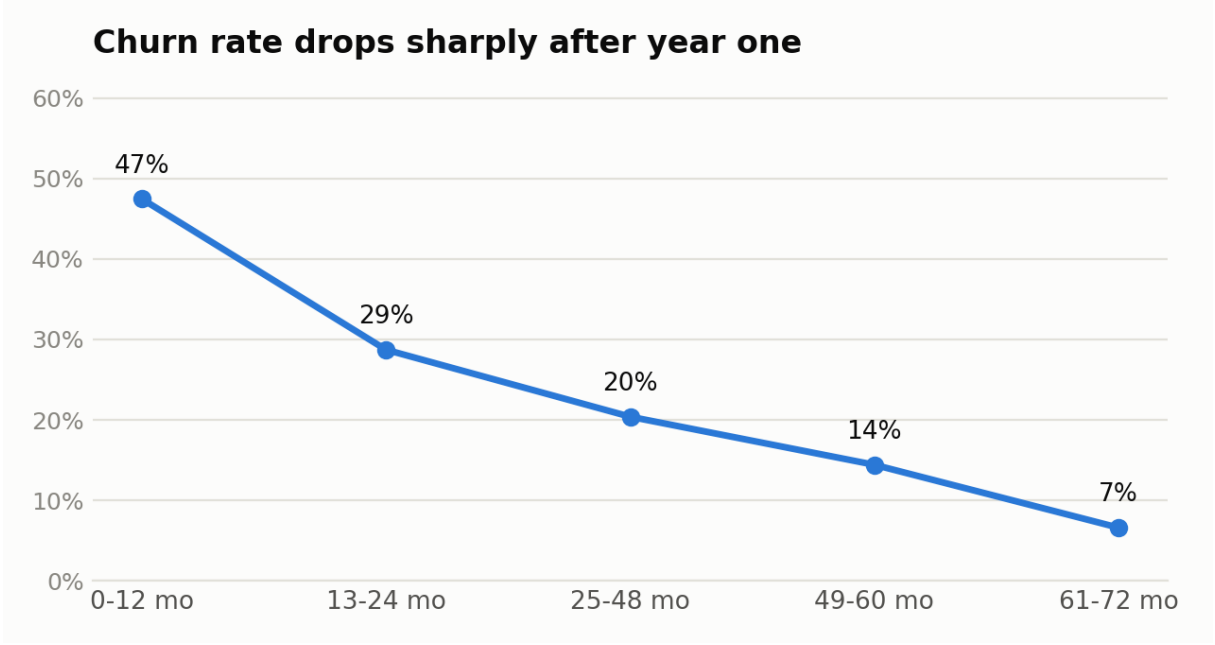


Fig. 2 — Query 3: churn rate by tenure bucket. Retention risk is front-loaded into year one.

The two compound further: fiber-optic customers on month-to-month contracts *without* a tech-support bundle churn at 57.5% — 2.2x the company-wide average — and this segment alone (1,796 customers) carries $1,057,989 of annualized revenue at risk. Isolating the bundle question on fiber customers specifically (query 6) shows the gap holds even controlling for contract type: 55.0% churn with no security/support bundle vs. 24.5% with one.

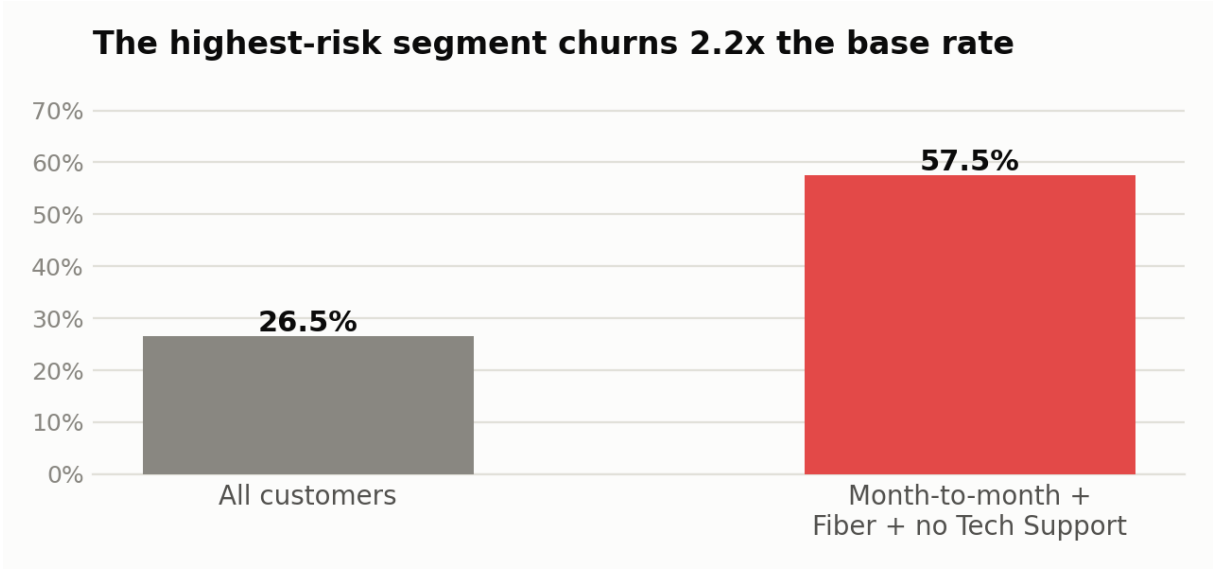


Fig. 3 — Query 9: the highest-risk segment vs. the company-wide base rate.

The AI checkpoint

Feeding the query results (not the raw dataset) to an AI assistant surfaced three competing explanations: price sensitivity among high-bill customers, contract flexibility itself, and a missing support bundle. The first

was set aside — churners do pay more on average ($74.44 vs $61.27/mo) but that gap is confounded with fiber pricing and mostly disappears once bundle status is controlled for in query 6. The bundle-gap explanation is what survived cross-checking against the query output, and it's the one this report leads with.

Recommendation

Target the 1,796-customer segment first: fiber-optic, month-to-month, no tech-support bundle. A retention offer that converts even a third of this segment onto a support bundle or a longer contract term would put a meaningful share of the $1,057,989 in segment-level annual revenue at risk back on a retention path — and it's a narrowly-targeted enough segment to pilot before rolling out company-wide.

RESUME BULLET, READY TO USE

“Led end-to-end churn analysis on 7,043 telecom customers — authored a BRD/FRS, wrote 10 SQL queries, and identified a customer segment churning 2.2x the base rate and representing $1.06M of annual revenue at risk.”

What's in this package

File	What it is
churn_analysis.sql	All 10 SQL queries behind every figure in this report — run it yourself against the dataset.
run_analysis.py	Python script: loads the CSV into SQLite, runs the SQL, generates the 3 charts above.
This report (PDF)	The BRD, FRS, findings, and recommendation — the write-up you'd actually hand to a stakeholder.

Appendix — SQL excerpt

```
-- Query 9: highest-risk segment vs. base rate
SELECT COUNT(*) AS customers_in_segment,
  ROUND(100.0*SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END)/COUNT(*), 2) AS churn_rate_pct
FROM customers
WHERE Contract = 'Month-to-month'
  AND InternetService = 'Fiber optic'
  AND TechSupport = 'No';
```

Full query file (10 queries, numbered and commented) included alongside this report.